

Light-driven motility in microalgae

Sujeet Kumar Choudhary & Ahmad Badr

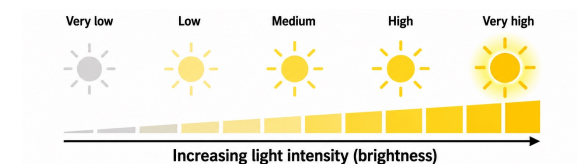
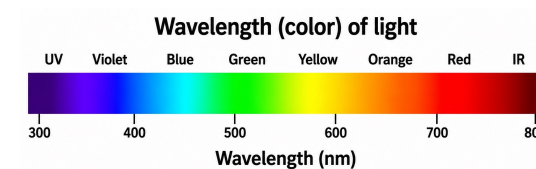
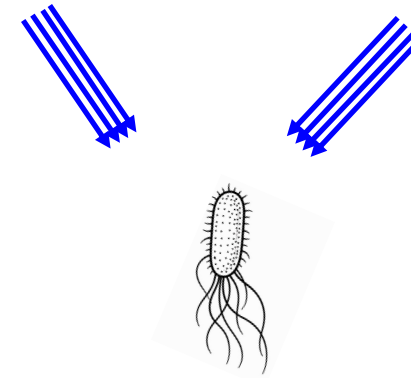
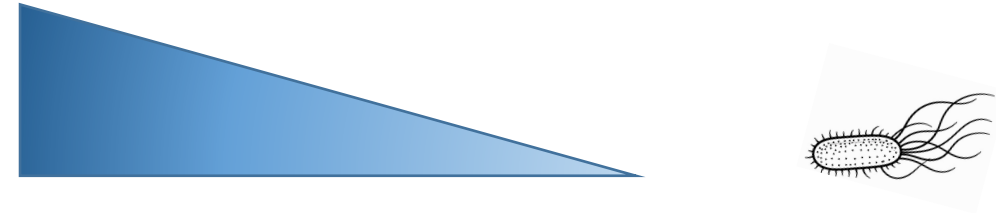
Mediterranean Institute of Advanced Study, Mallorca, Spain

Université de Bordeaux, France

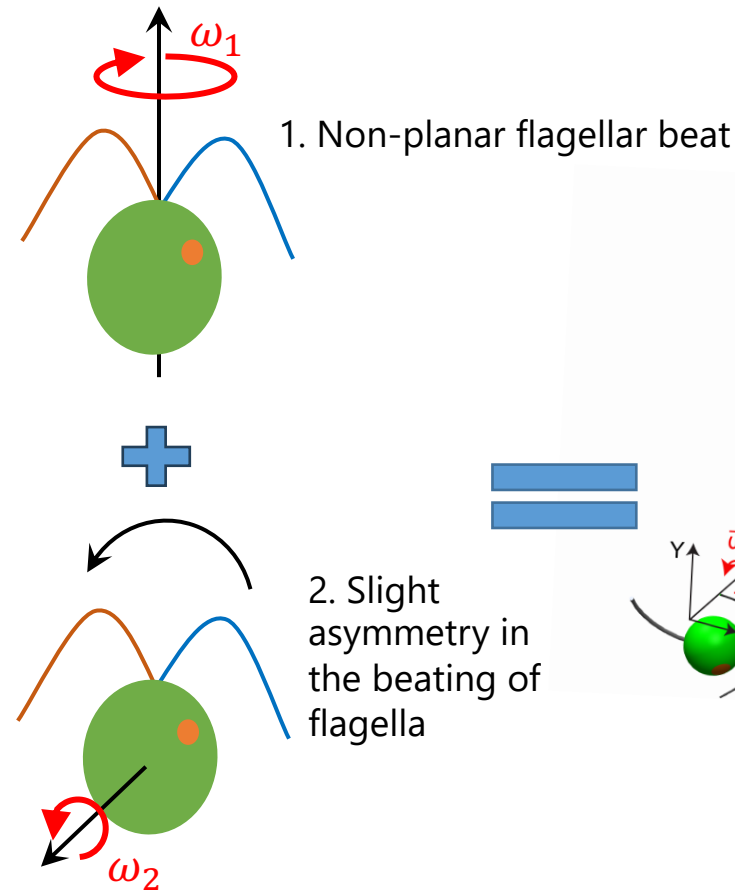
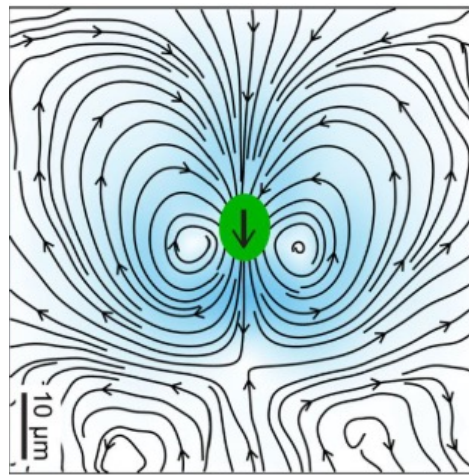
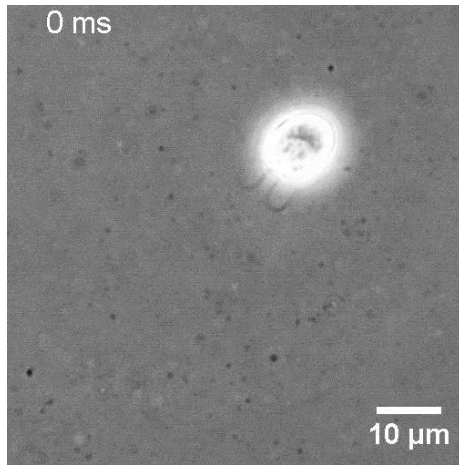
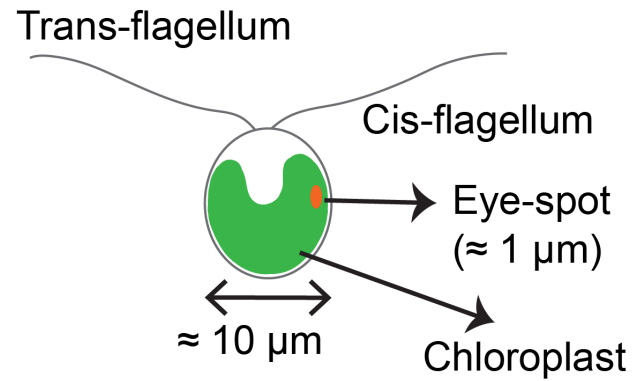
How do microscopic organisms process the light to adapt their motility?

A cell with no brain can still “decide” where to go
-Is it simple physical and biochemical rules?

- How does a single cell detect a light gradient in the environment?
- Can a microorganism “prefer” one direction over another?
- How does the motility depend on the wavelength and intensity of the light?

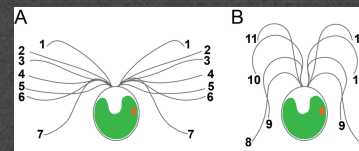
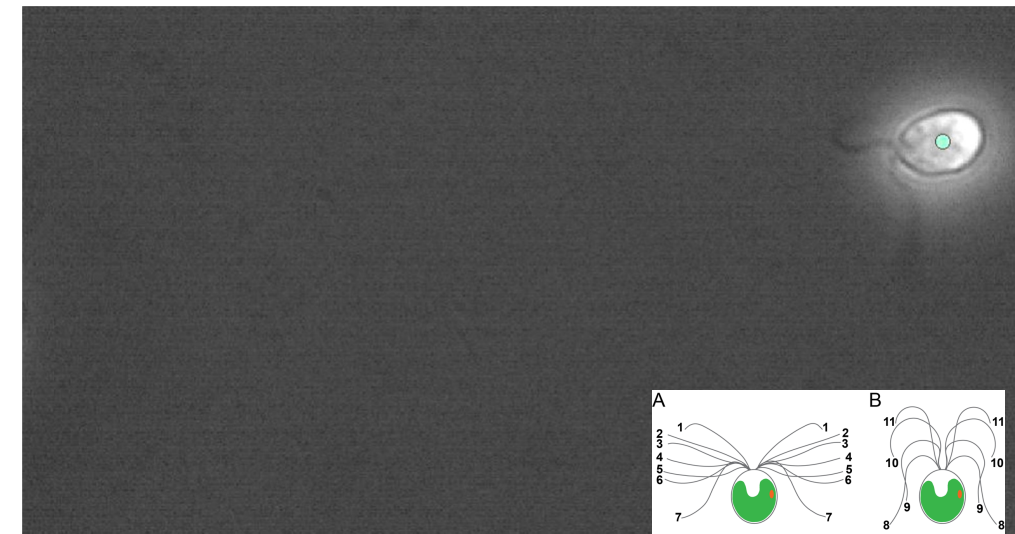


Meet *Chlamydomonas*



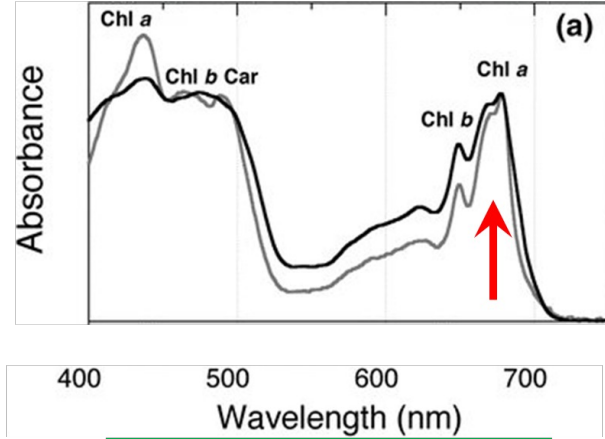
Flagellar beat frequency
~ 50 Hz

Cell speed
~ 100 $\mu\text{m/s}$



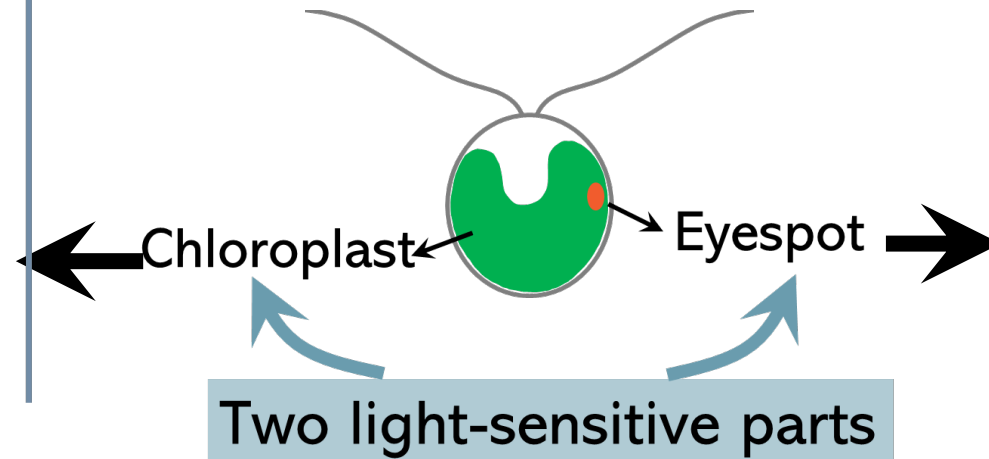
Chlamydomonas reinhardtii (CR)

Photosynthetically Active Radiation (PAR)

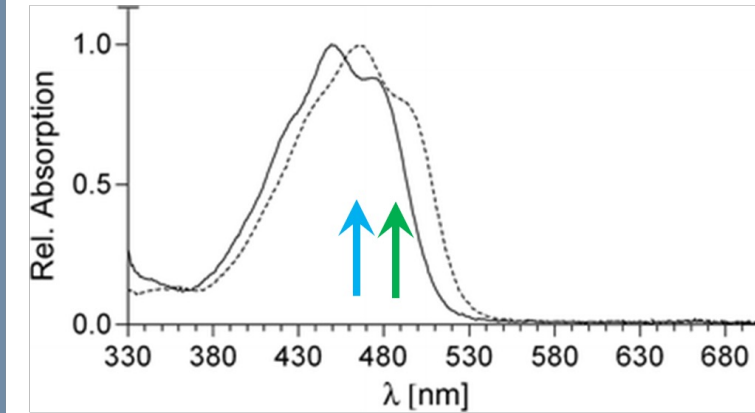


Chlorophyll spectra

Ueno et al., 2019

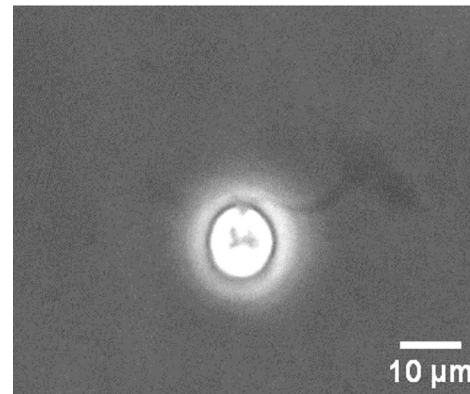


Phototactic light (Blue & Green)

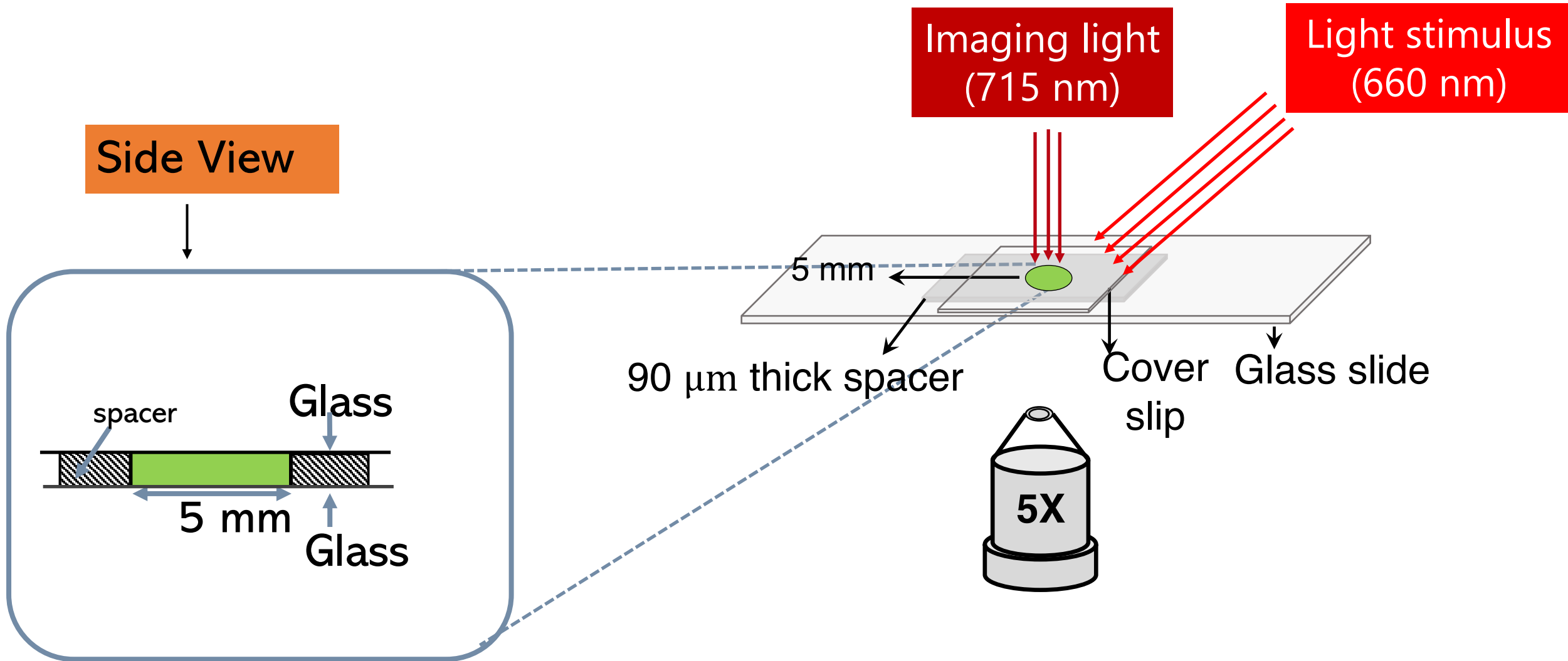


Eyespot spectra

Kreimer G., 2009

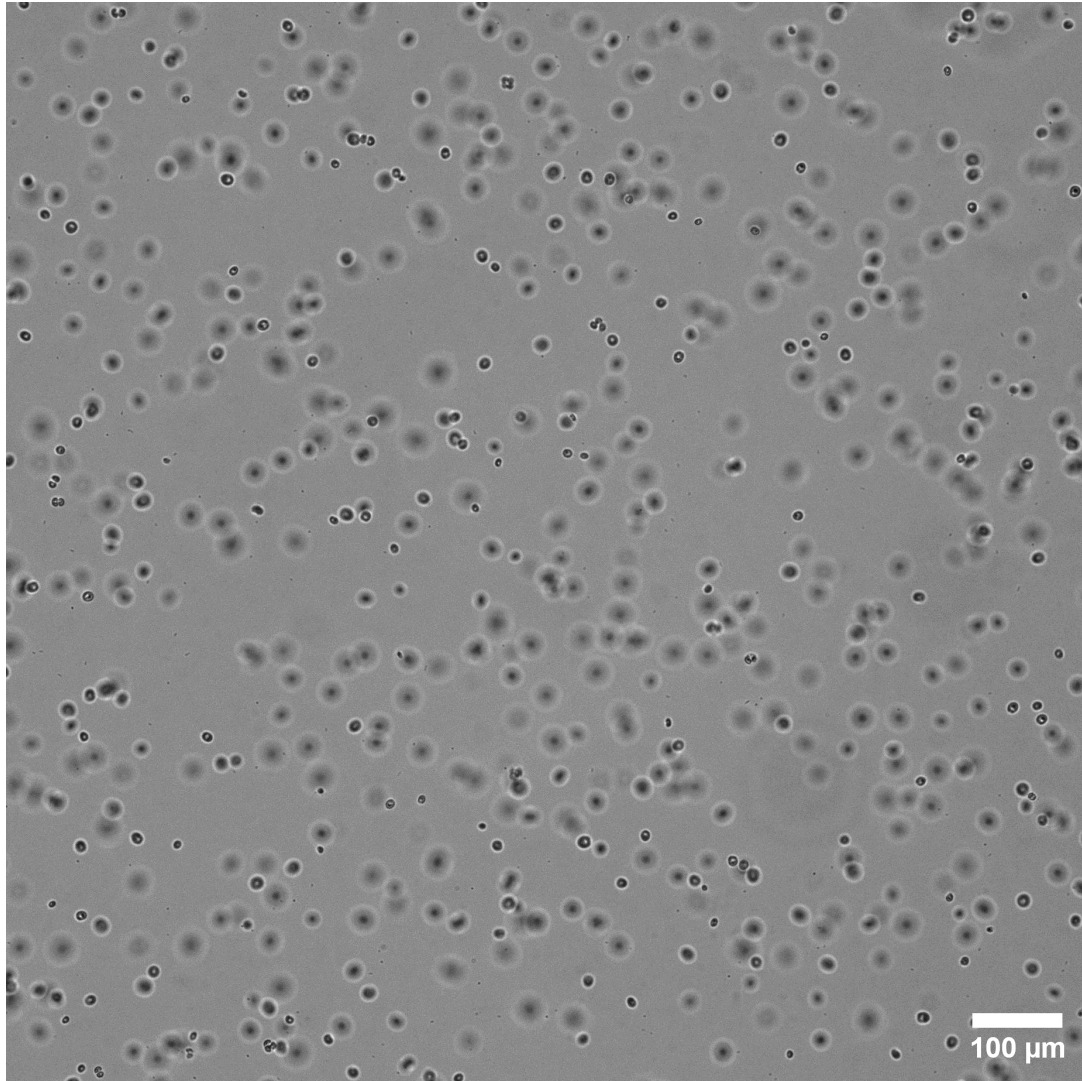


Experimental Setup: Photokinesis

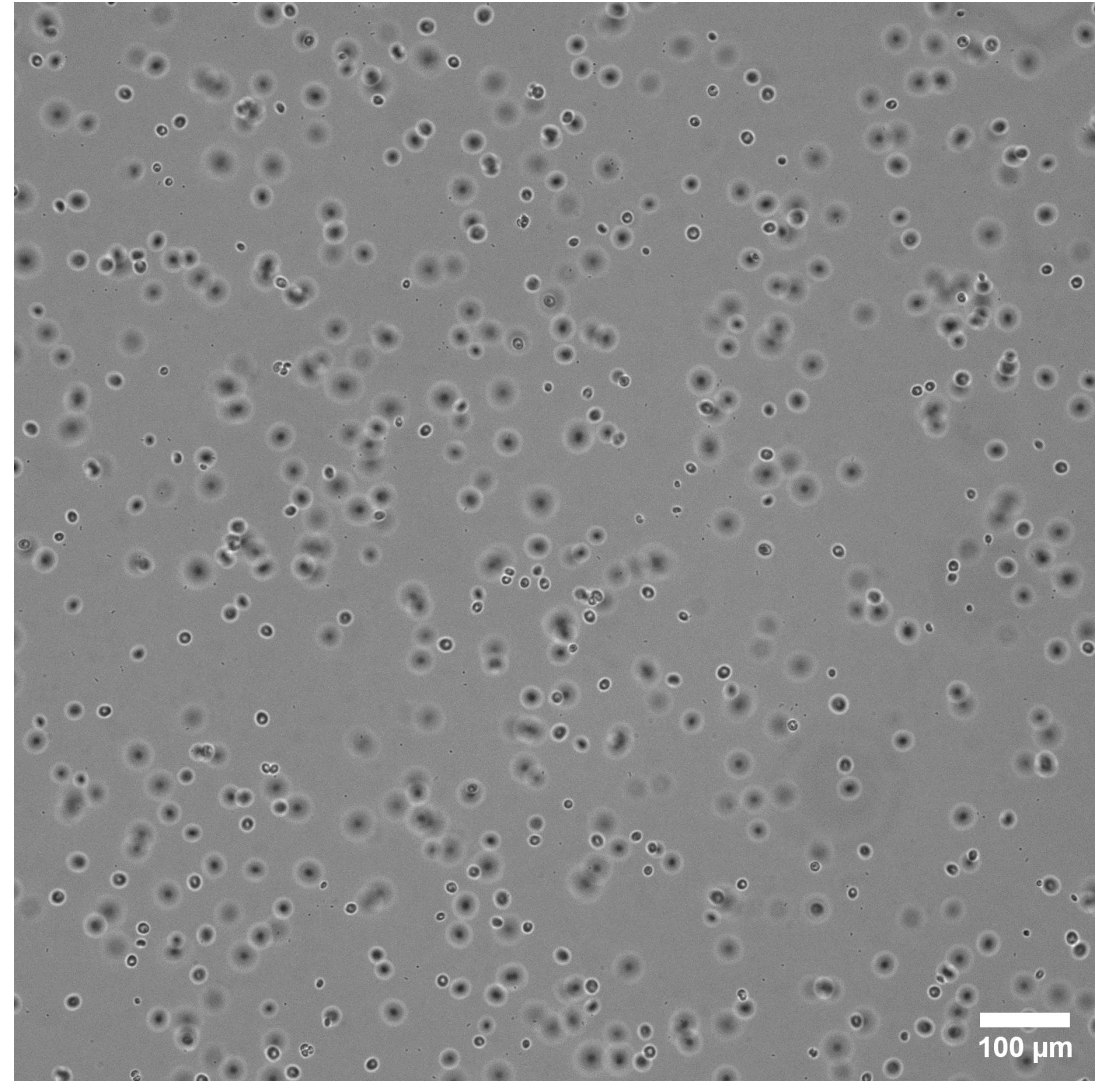


Photokinesis in *Chlamydomonas reinhardtii* cells

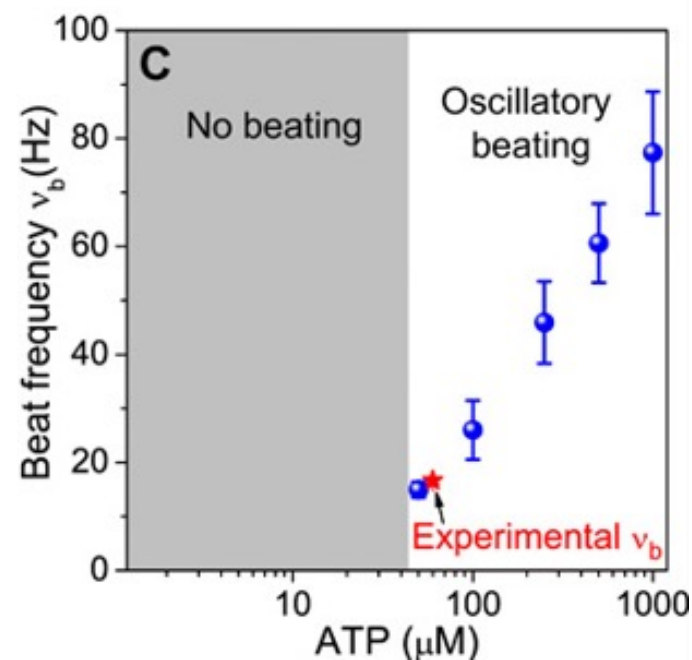
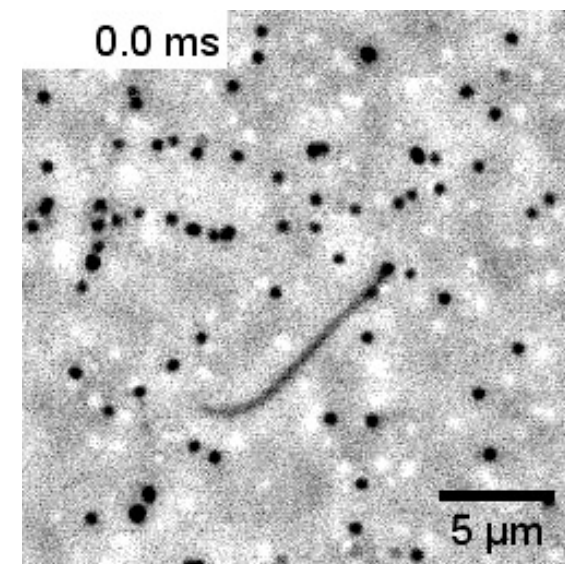
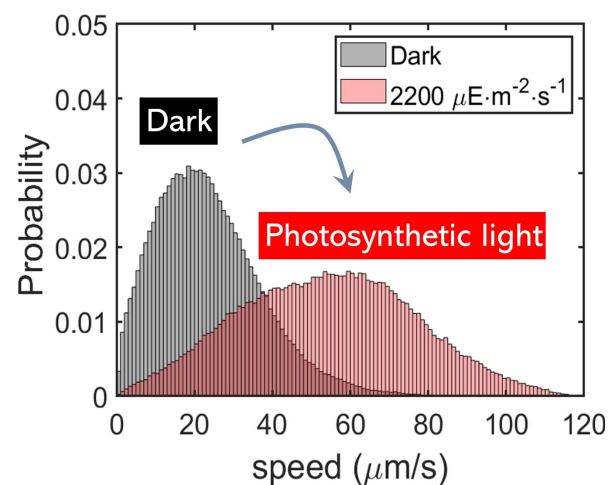
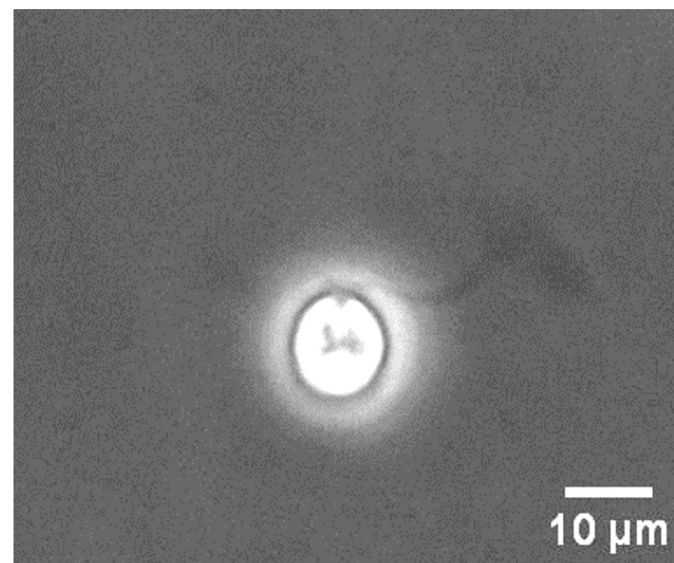
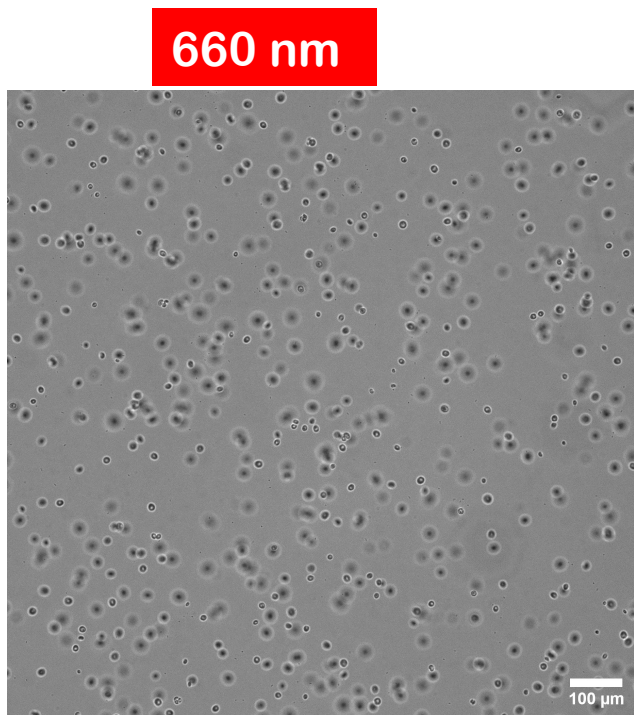
Dark



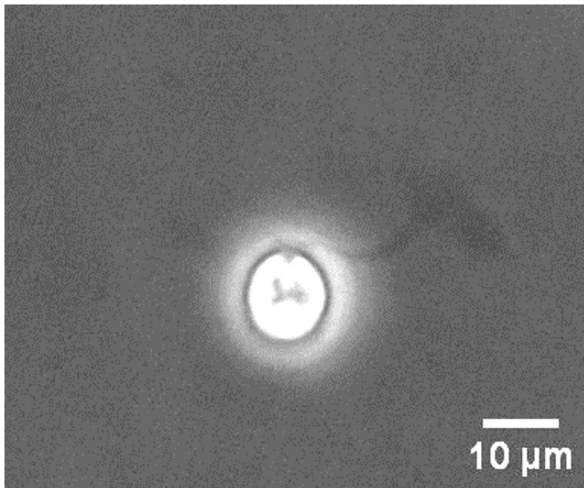
660 nm



Mechanism of photokinesis



Mechanism of photokinesis



Energy production process

With PAR light

Photosynthesis in chloroplast (**Dominant Process**): **Produces oxygen, glucose, and ATP**

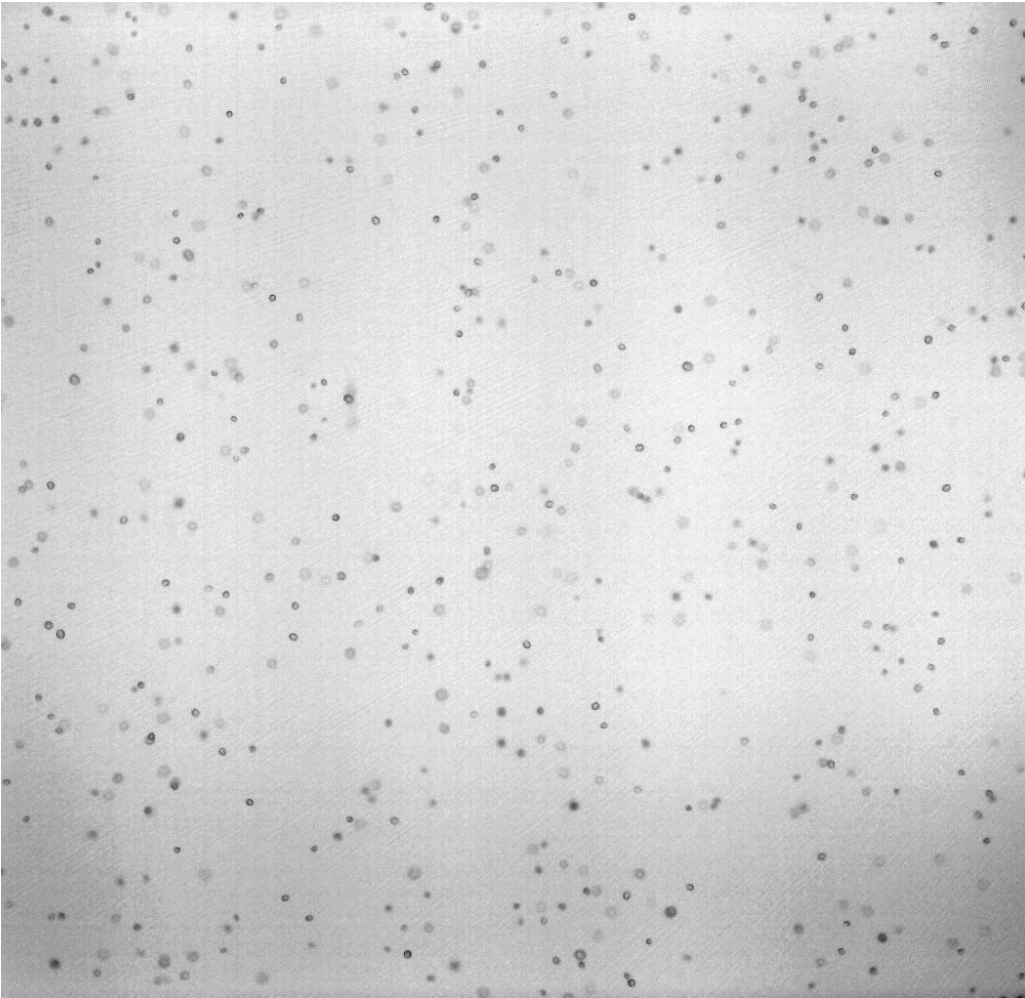
Cellular Respiration in mitochondria (**Reduced Process**)
consumes oxygen, glucose, and produces ATP

Without PAR light

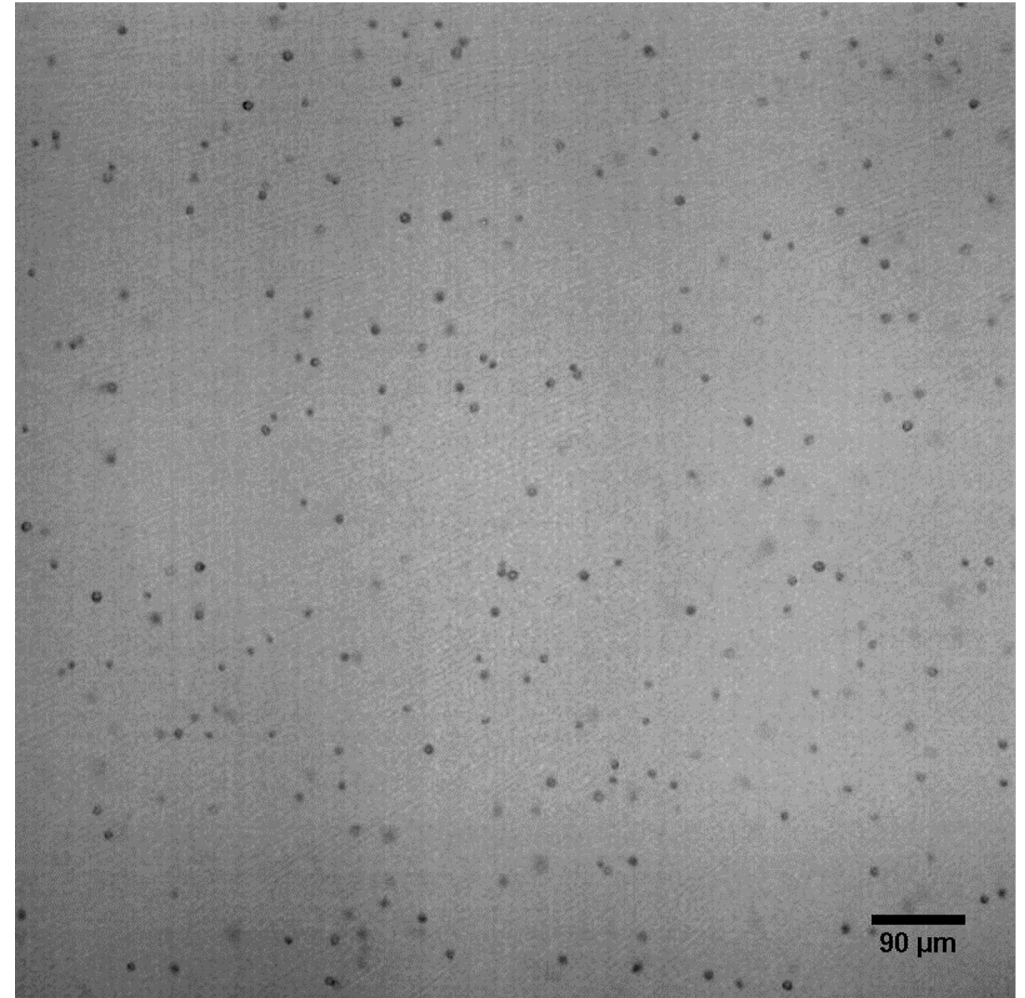
Cellular Respiration in mitochondria (**Dominant Process**):
consumes oxygen, stored glucose **and produces ATP**

Phototaxis in *Chlamydomonas reinhardtii* cells

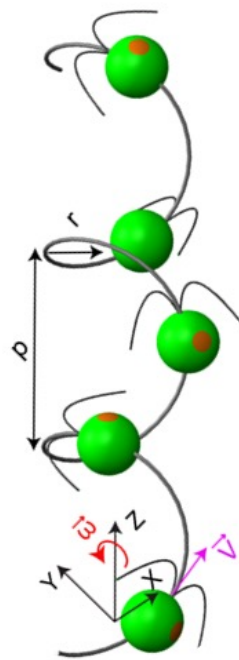
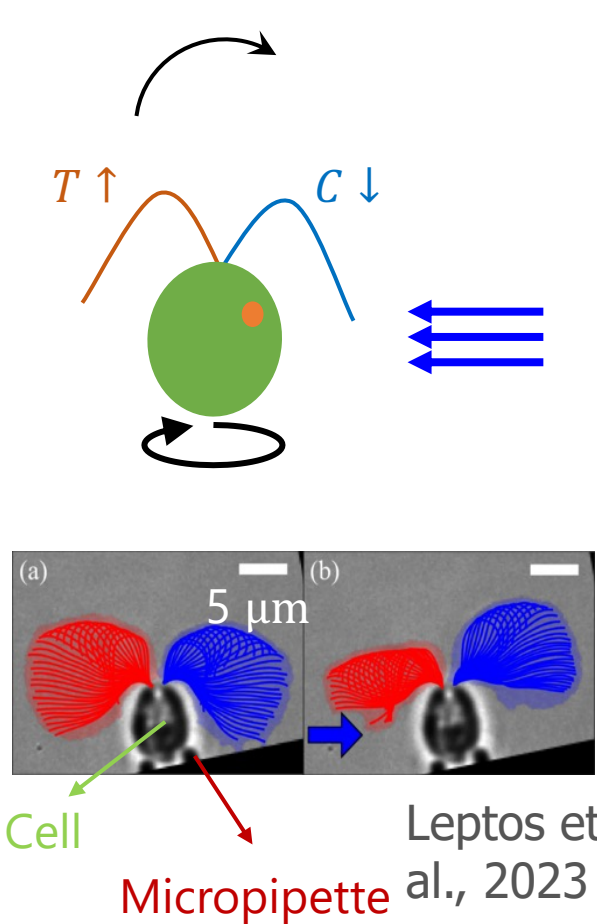
Dark



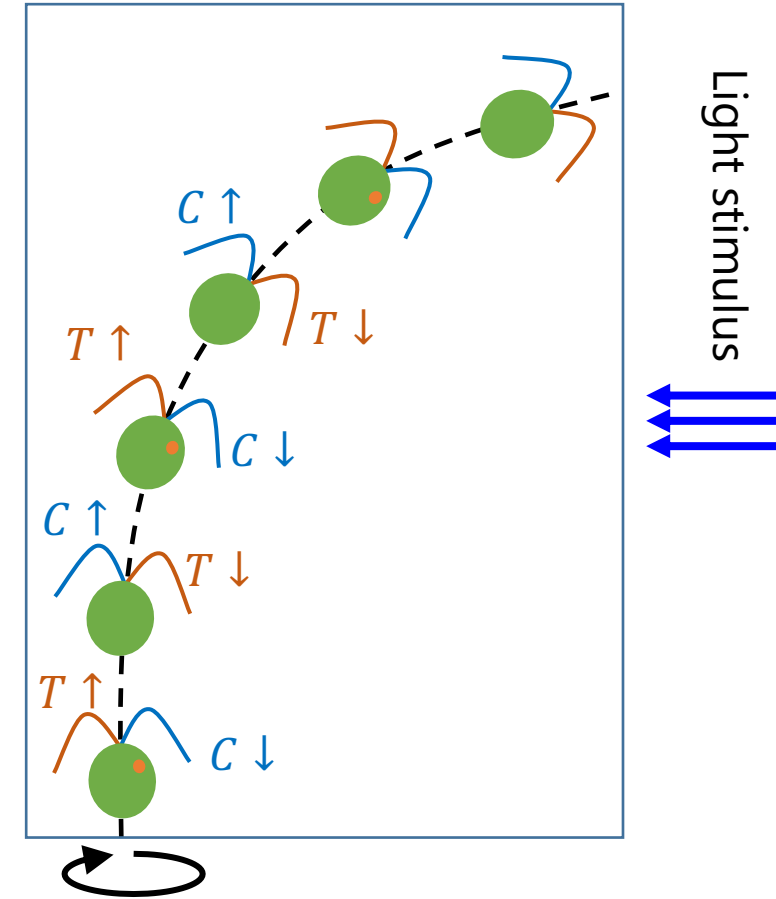
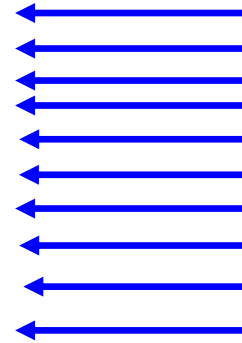
Blue/Green light



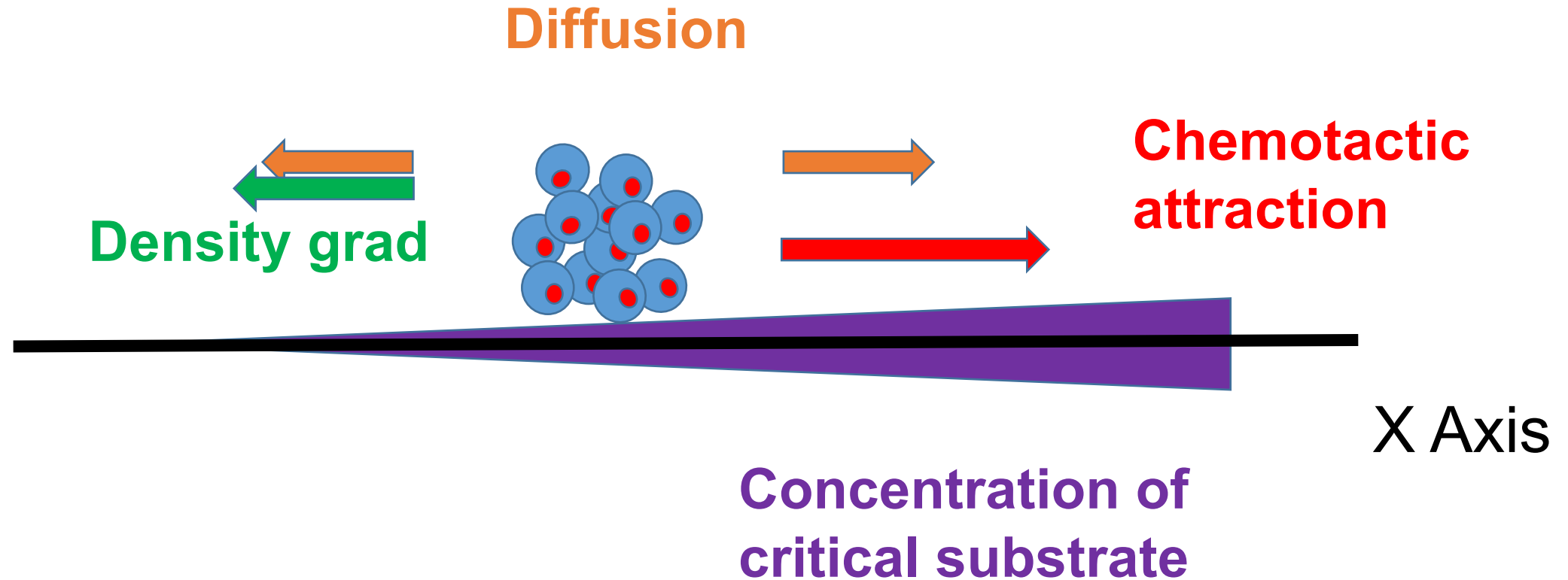
Mechanism to turn towards the light source



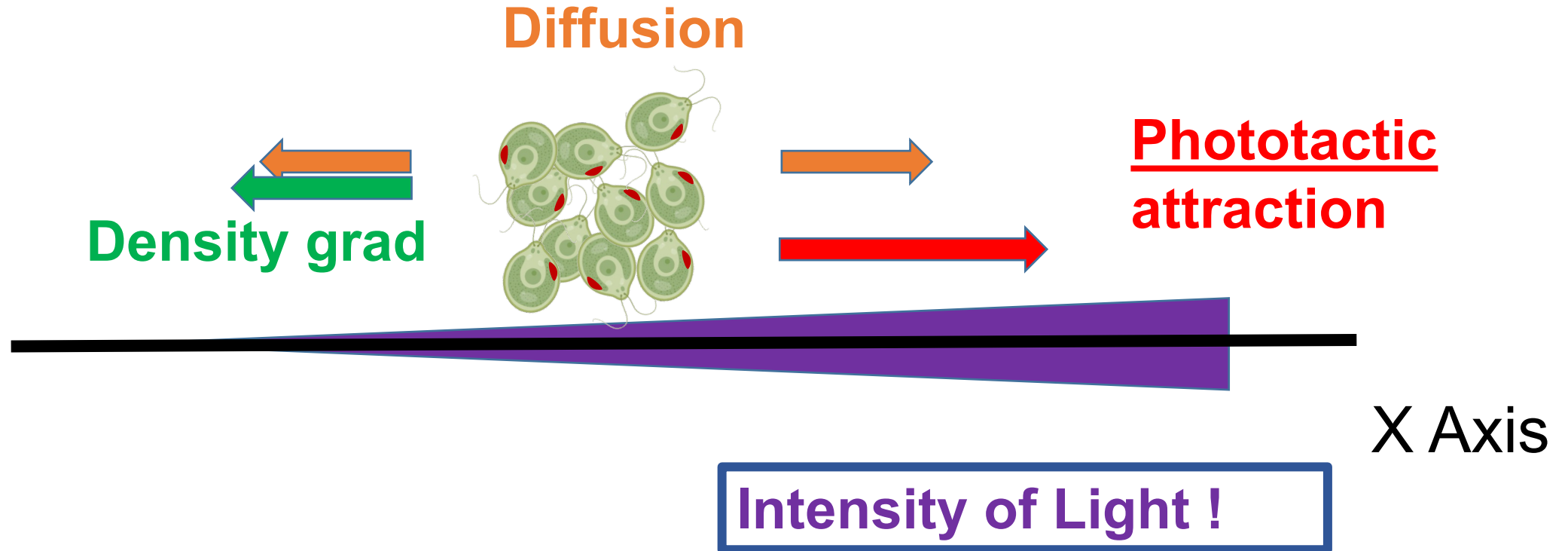
Light dark modulation



Model for Chemotaxis, Keller and Segel (1970)



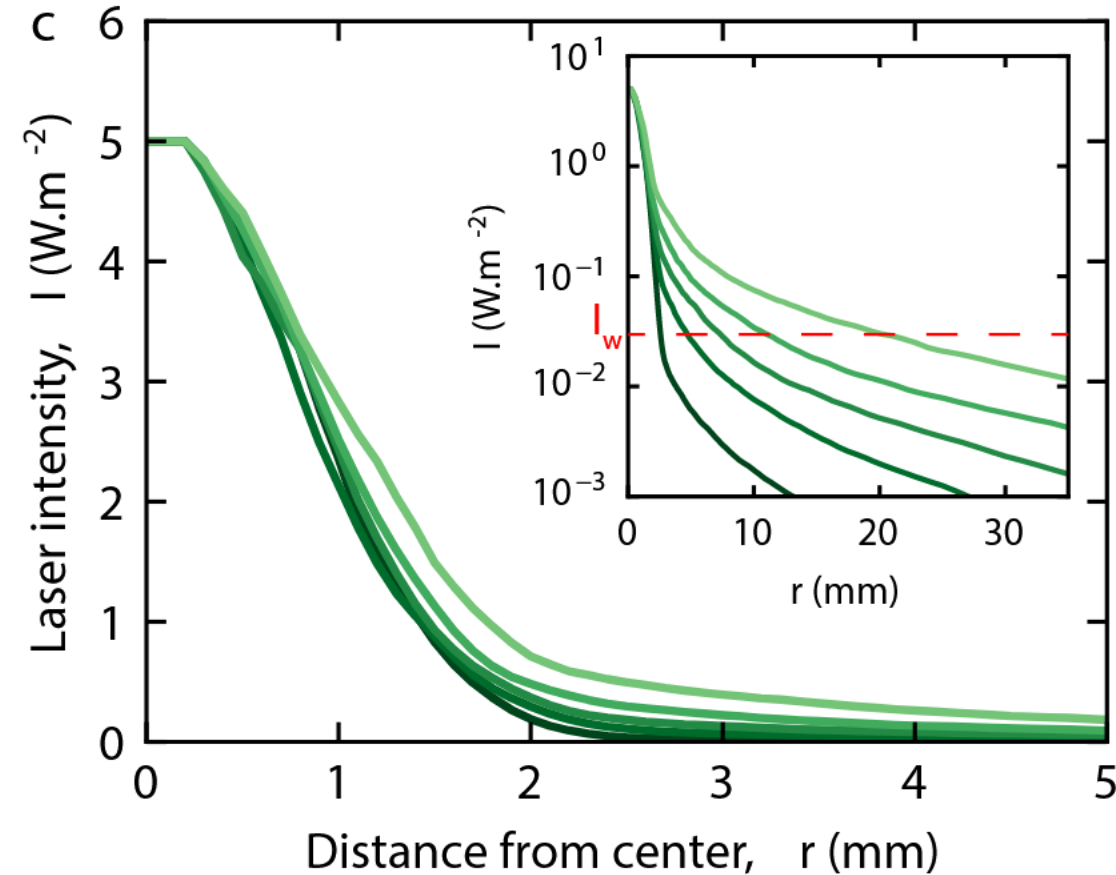
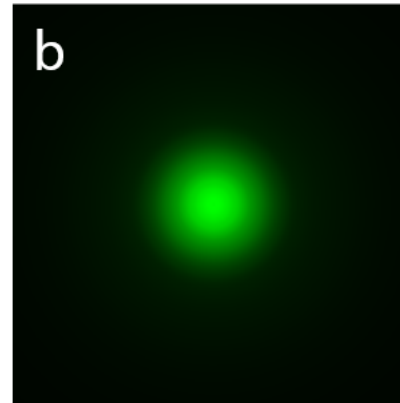
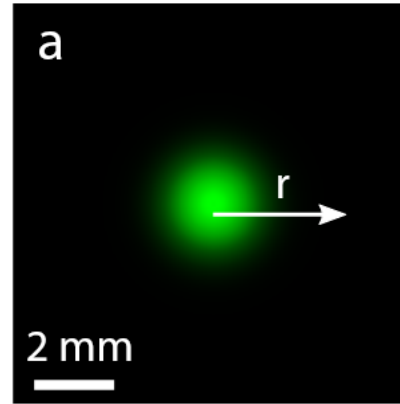
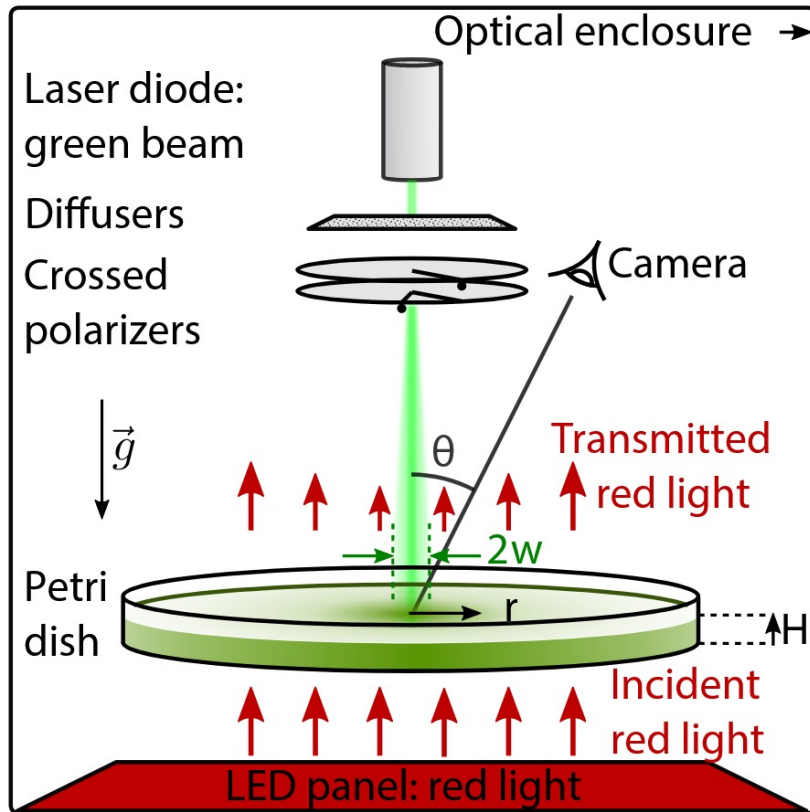
Model for Phototaxis



Swimming in an inhomogenous light field

Ramamonjy et al., PRL 2022

Green laser at 532 nm



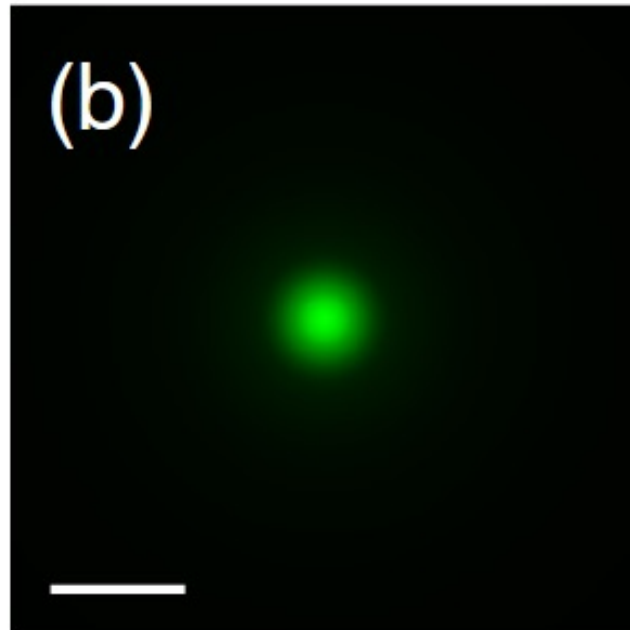
a & b : smallest and largest beam width

Red light : High pass at 650nm c: plot of Laser Intensity as a function of distance from the center

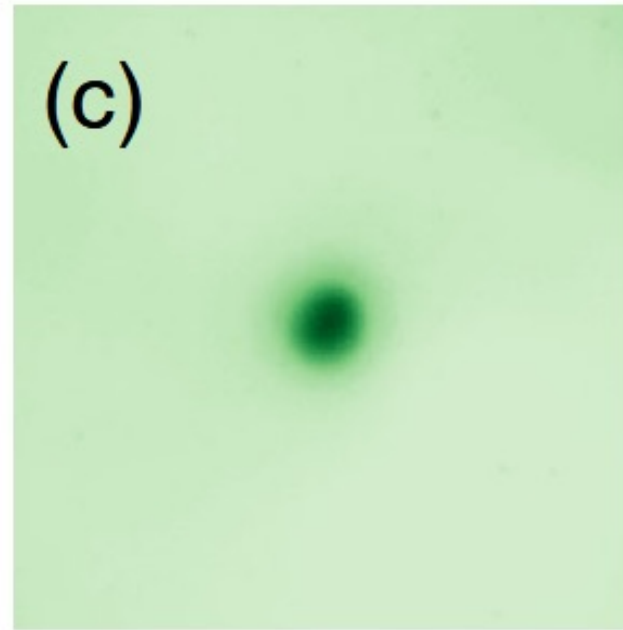
Stationary inhomogenous concentration field

Ramamonjy et al., PRL 2022

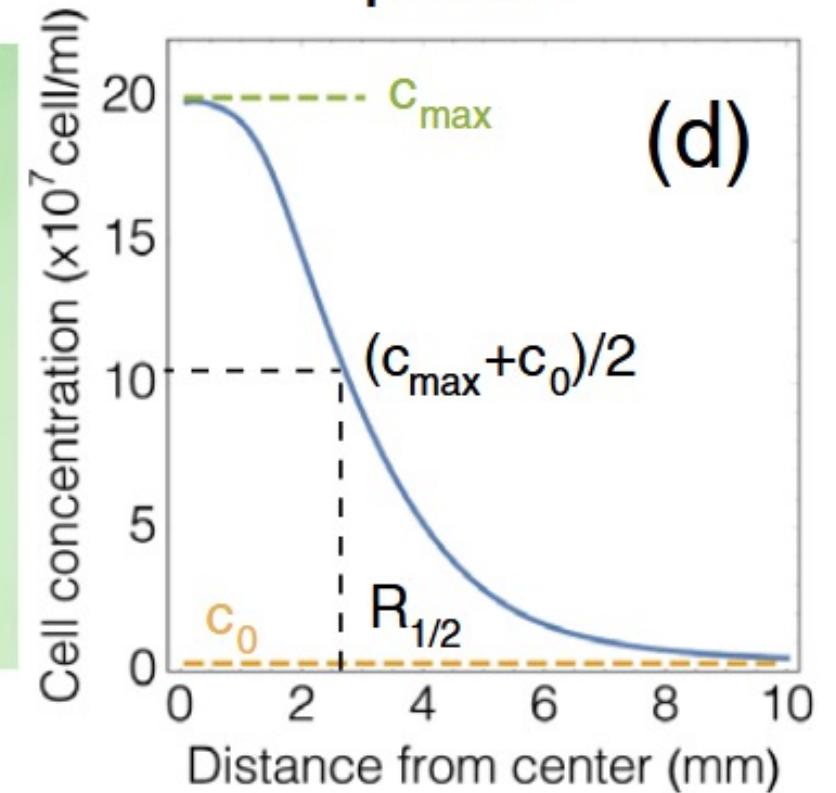
Luminous
landscape



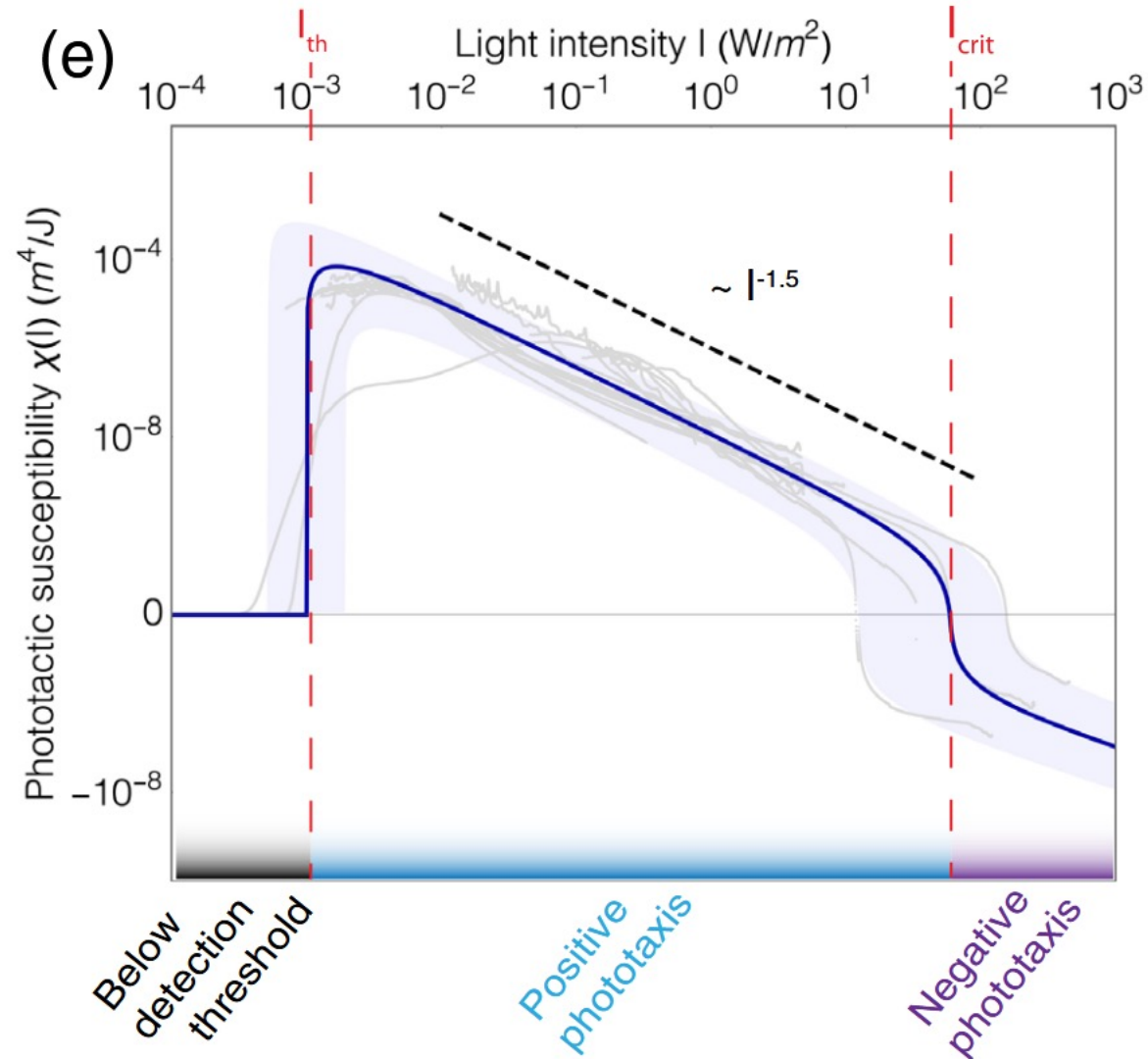
Algal
aggregation



Cell concentration
profile



Phototactic susceptibility $\chi(I)$

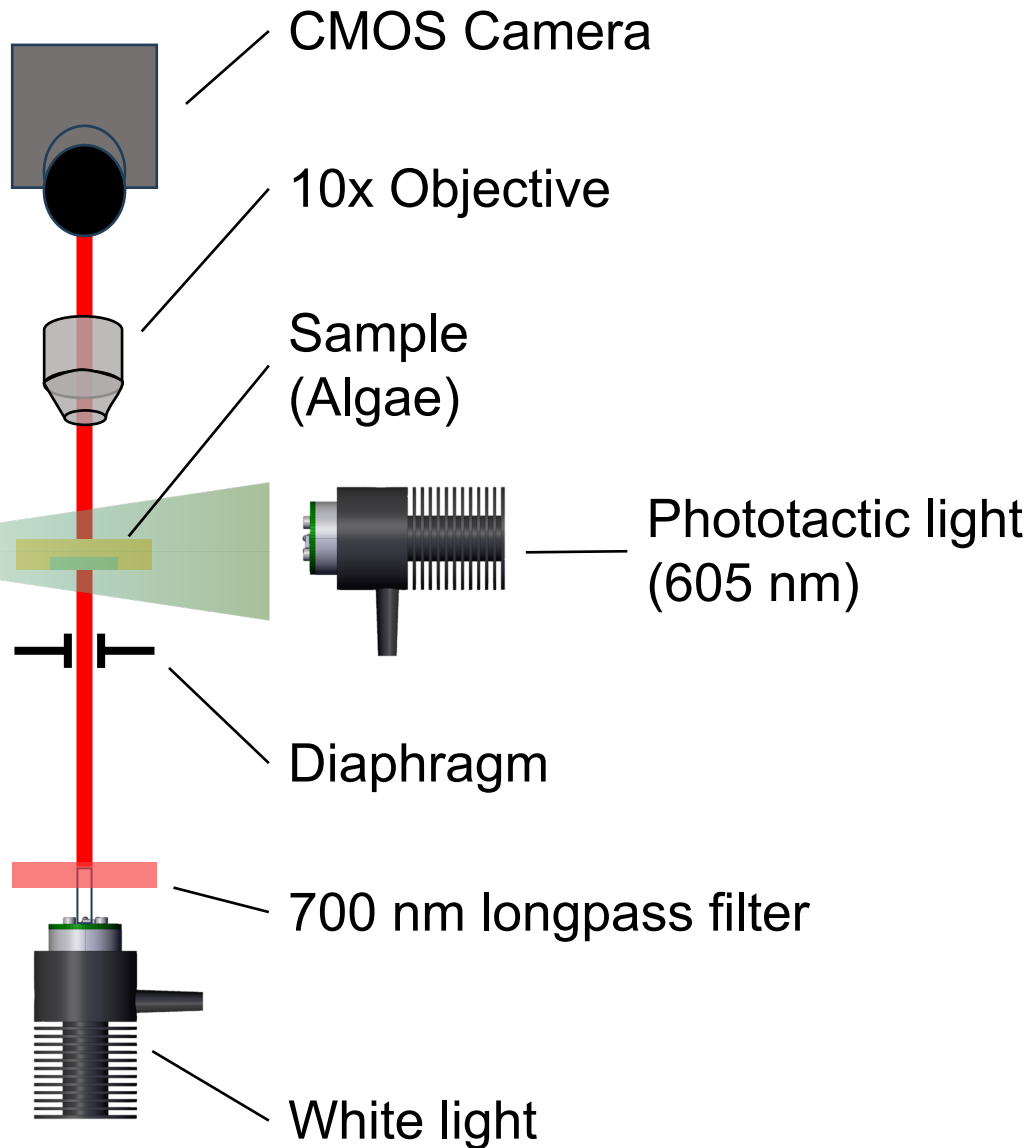


Identification of 3 different regimes :

- (1) no detection
- (2) Positive phototaxis
- (3) Negative phototaxis

Switches at $I_{\text{threshold}}$ and I_{crit}

Phototaxis experiment



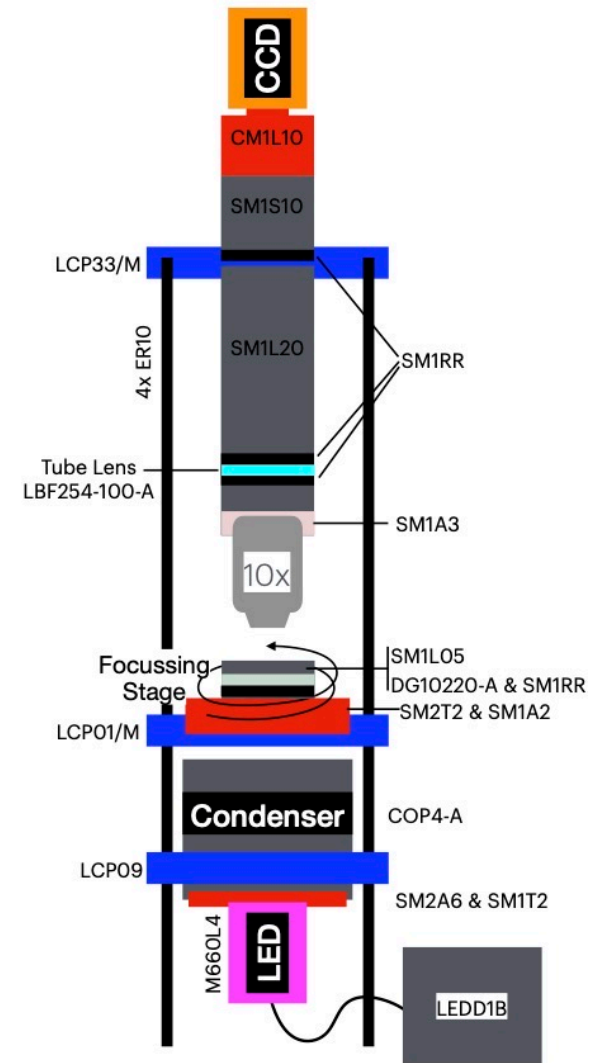
Record video :

- (1) with no phototactic light
- (2) with « weak » phototactic light
- (3) with « strong » phototactic light

Visualizing microscopic life



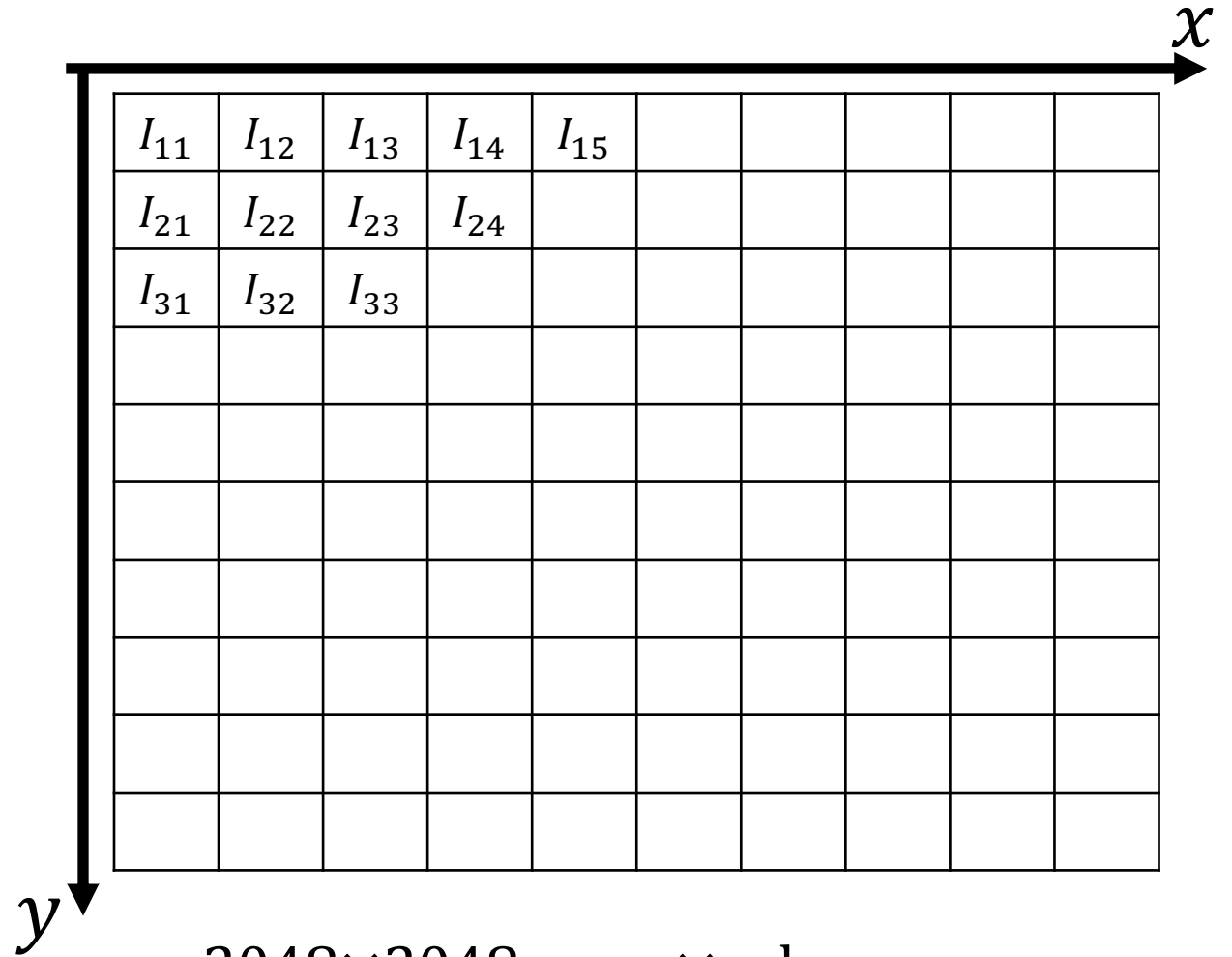
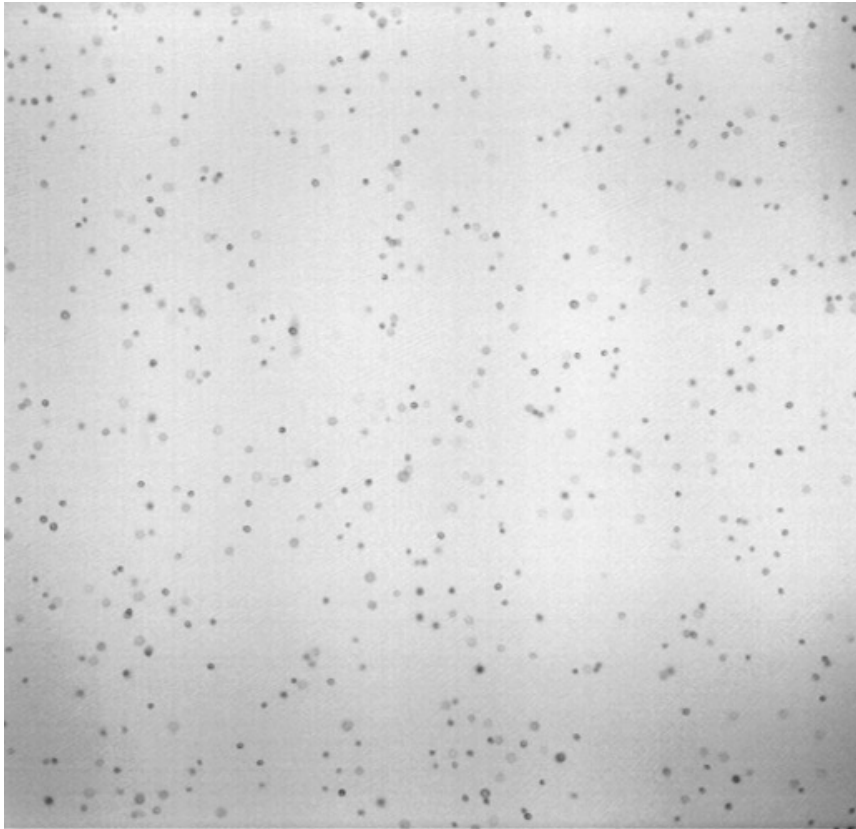
~35,000€



~1300€

Particle Tracking

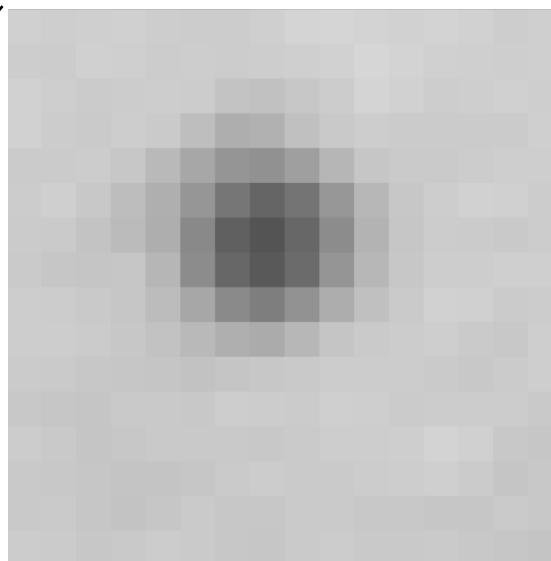
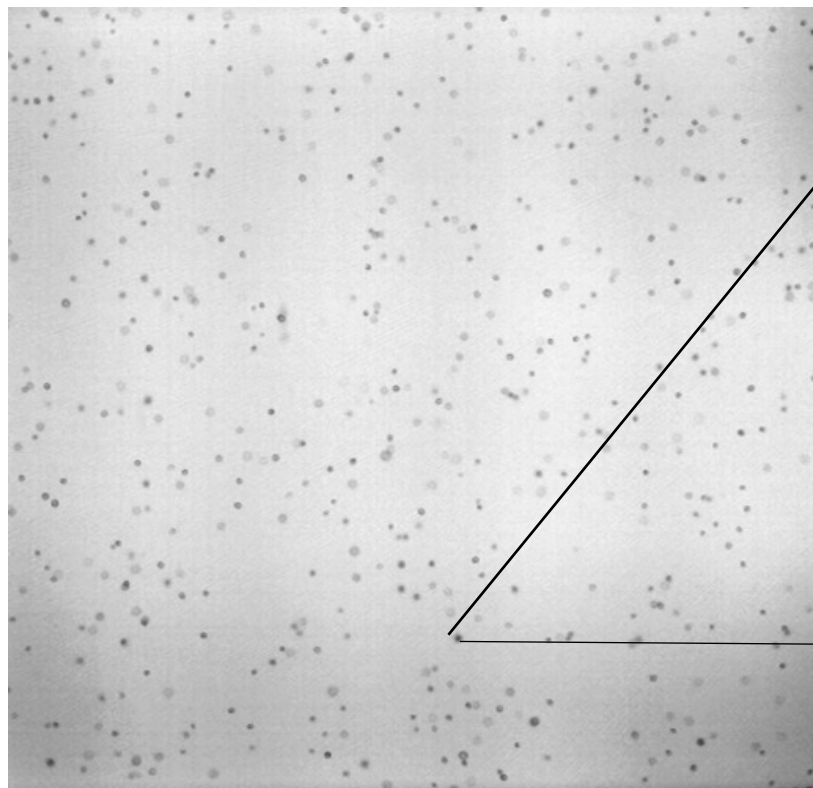
Reading an image



2048×2048, rows×column

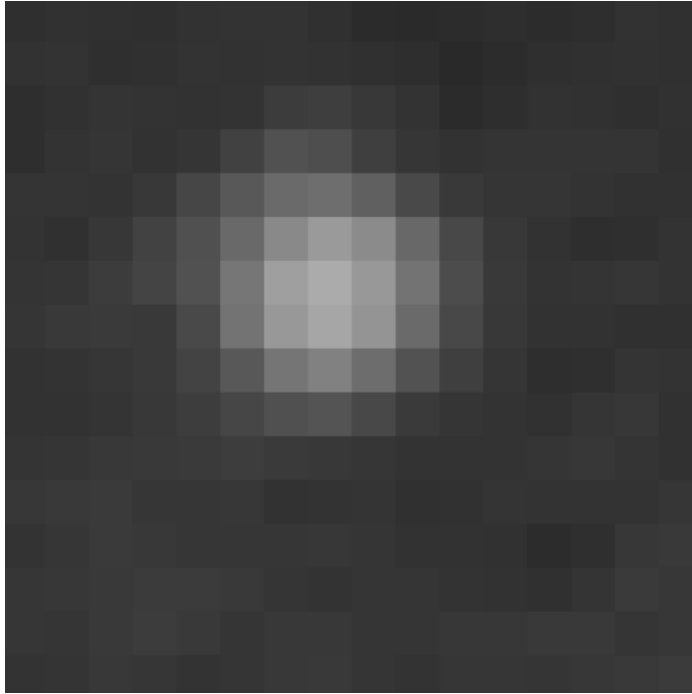
x, y coordinate of the pixel

Feature identification



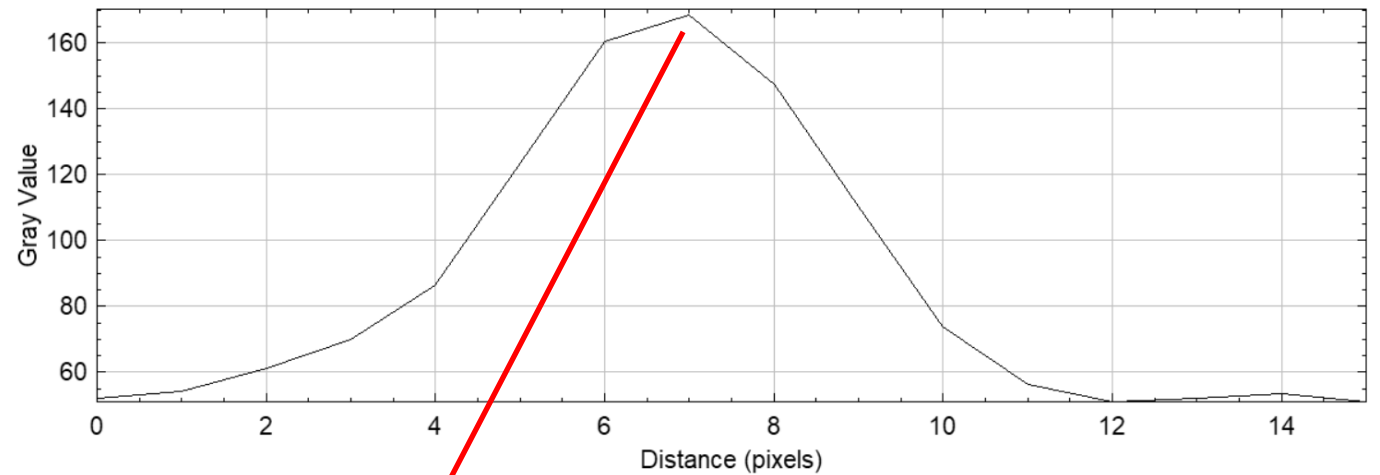
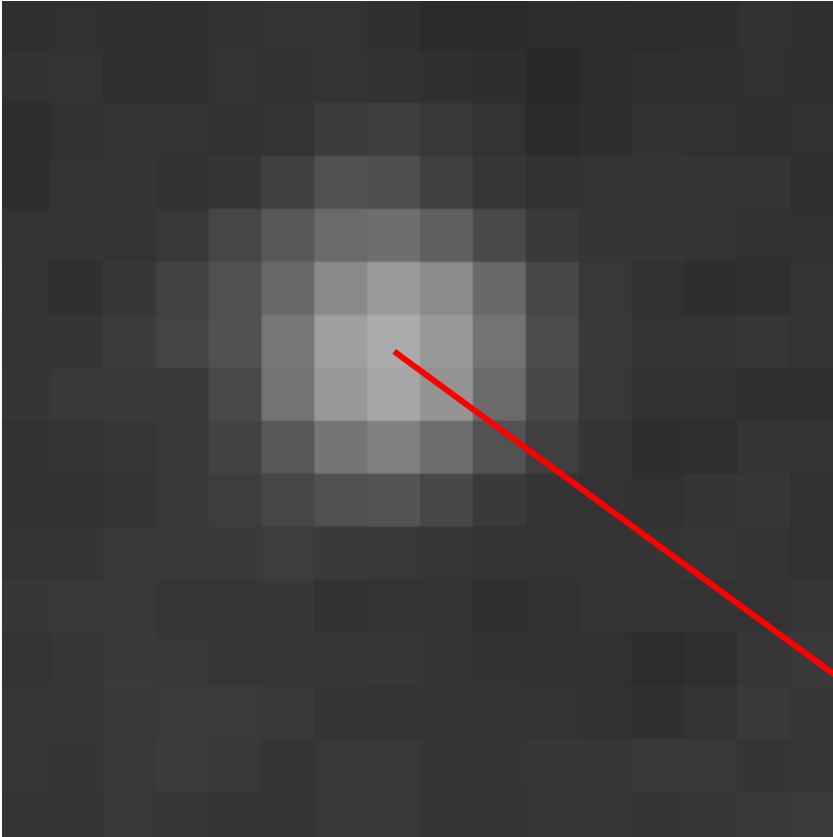
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204	207	200	189	175	150	118	101	116	151	183	199	205	209	208	204
203	202	195	187	174	137	96	84	103	140	179	198	204	203	201	204
202	198	197	198	182	140	103	89	107	149	183	199	205	205	207	207
205	204	201	198	188	167	138	126	146	173	192	202	209	208	203	204
205	205	203	199	194	185	175	171	183	197	202	204	206	202	200	206
203	202	199	198	197	194	197	199	201	204	204	204	202	200	203	206
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204	203	199	201	204	202	199	198	202	204	201	201	202	200	198	196

Inverting the image



47	49	48	47	50	51	51	48	43	42	44	46	44	46	50	48
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51	48	55	66	80	105	137	154	139	104	72	56	50	46	47	51
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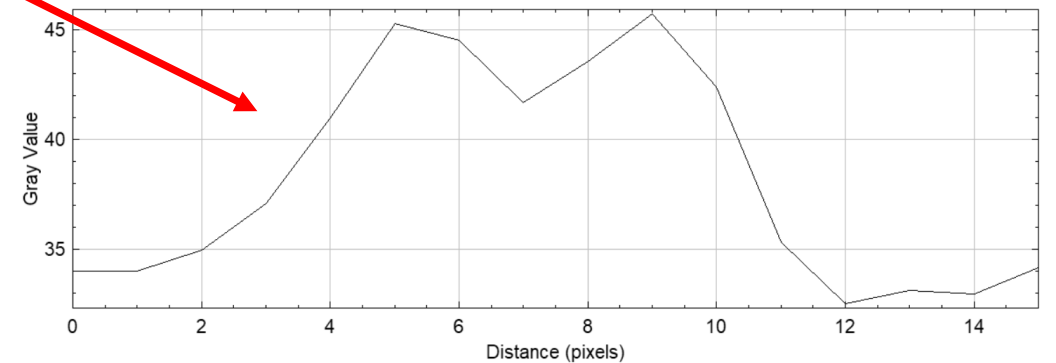
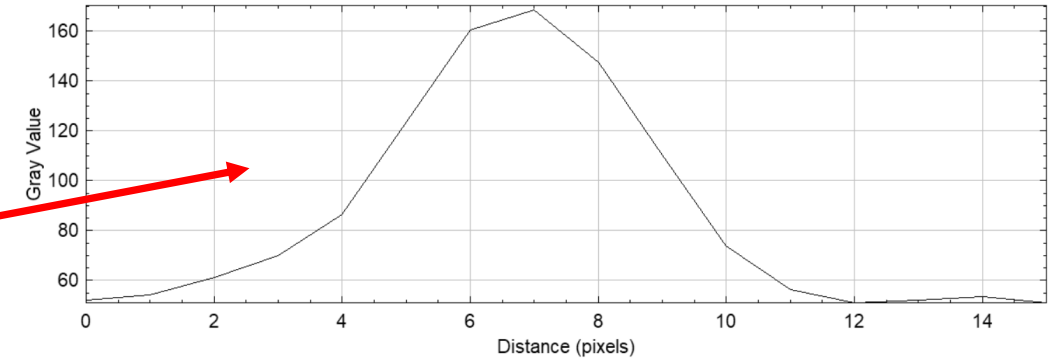
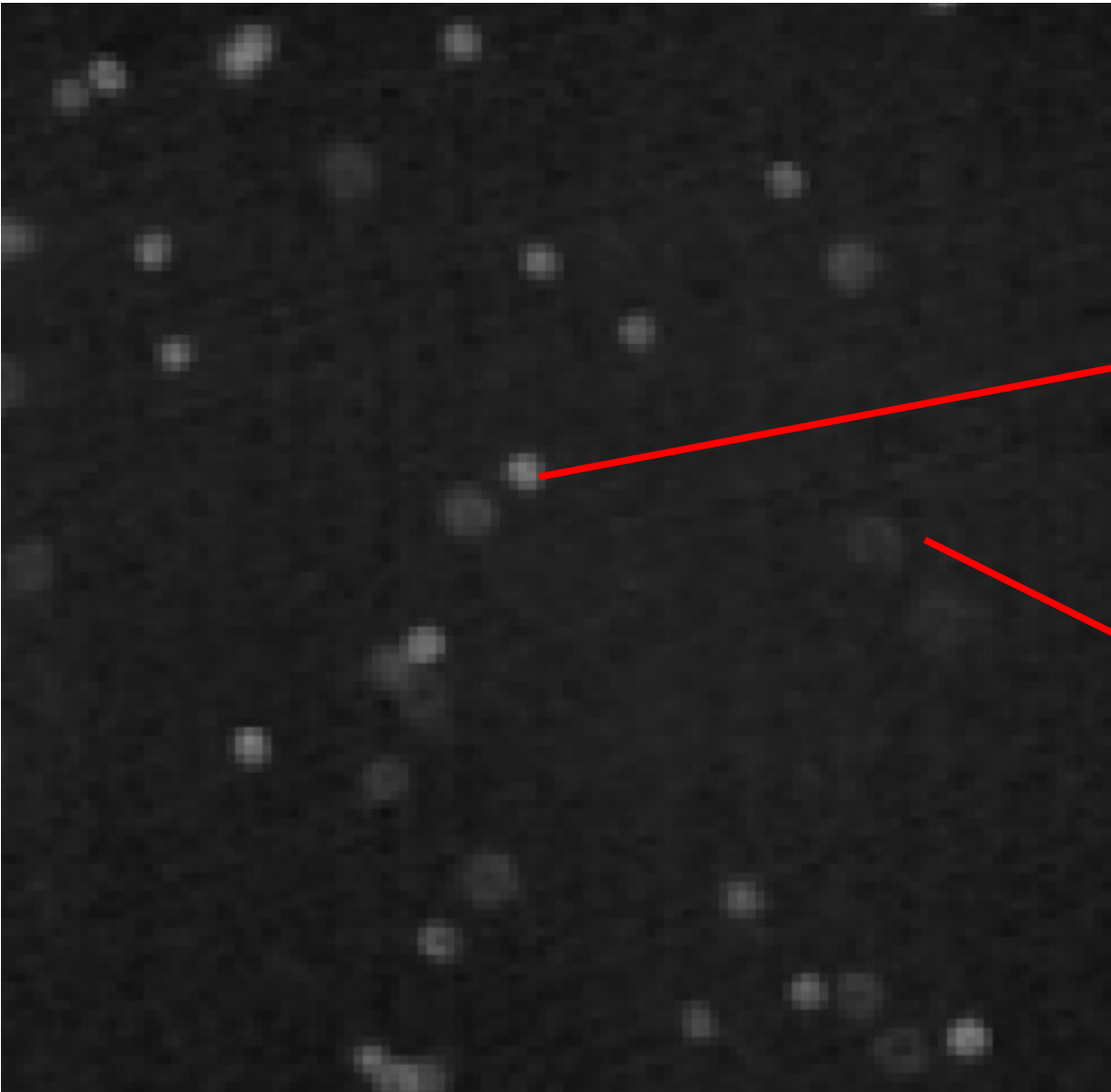
Intensity profile of the feature



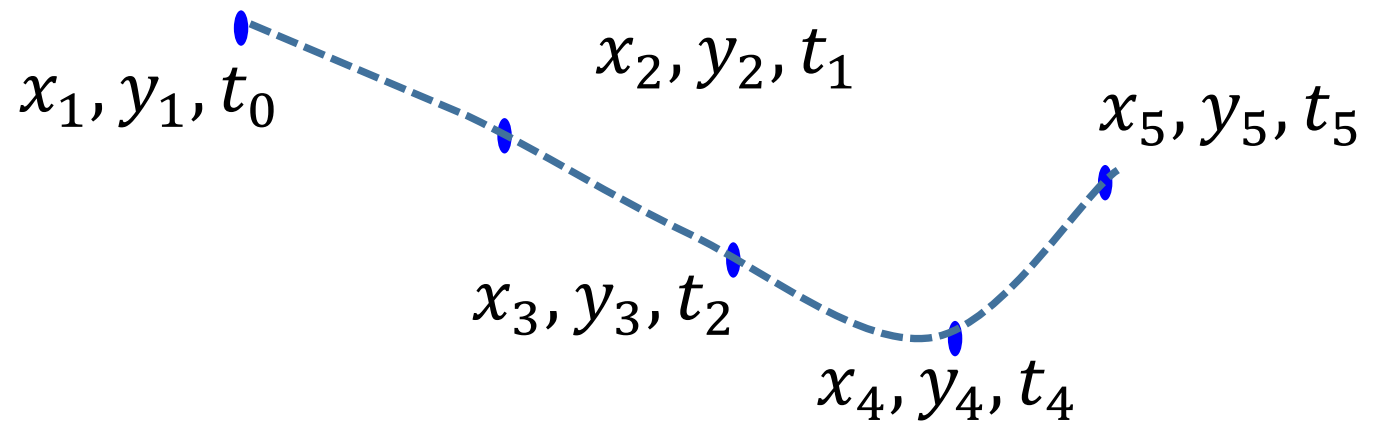
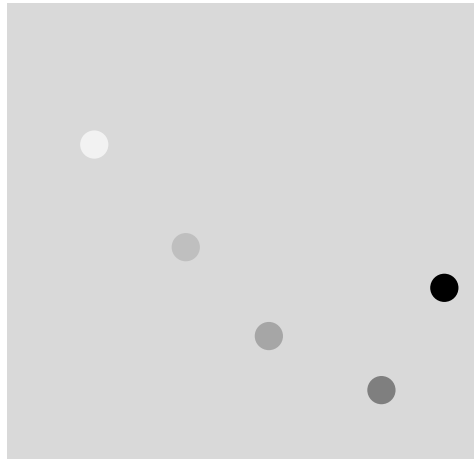
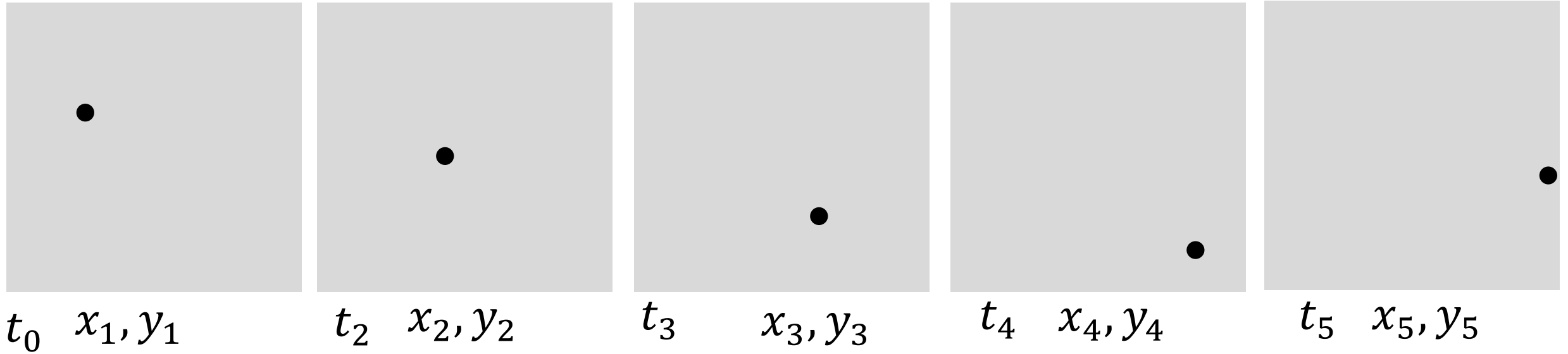
←————→
diameter of the particle in pixels

Position of the particle (x,y)

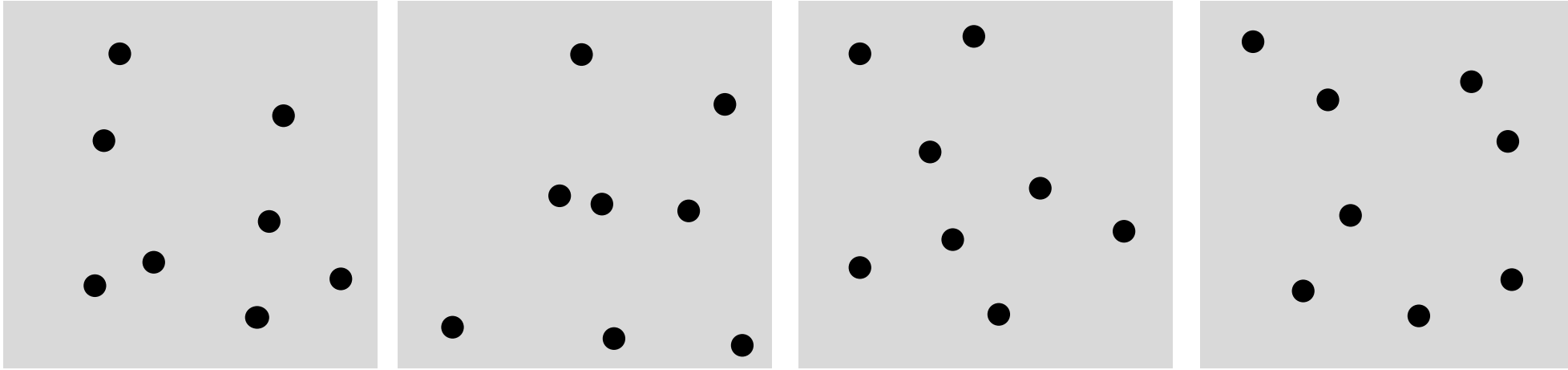
Intensity profile of the feature



Identifying the position of the particle in different frames and obtaining the trajectory

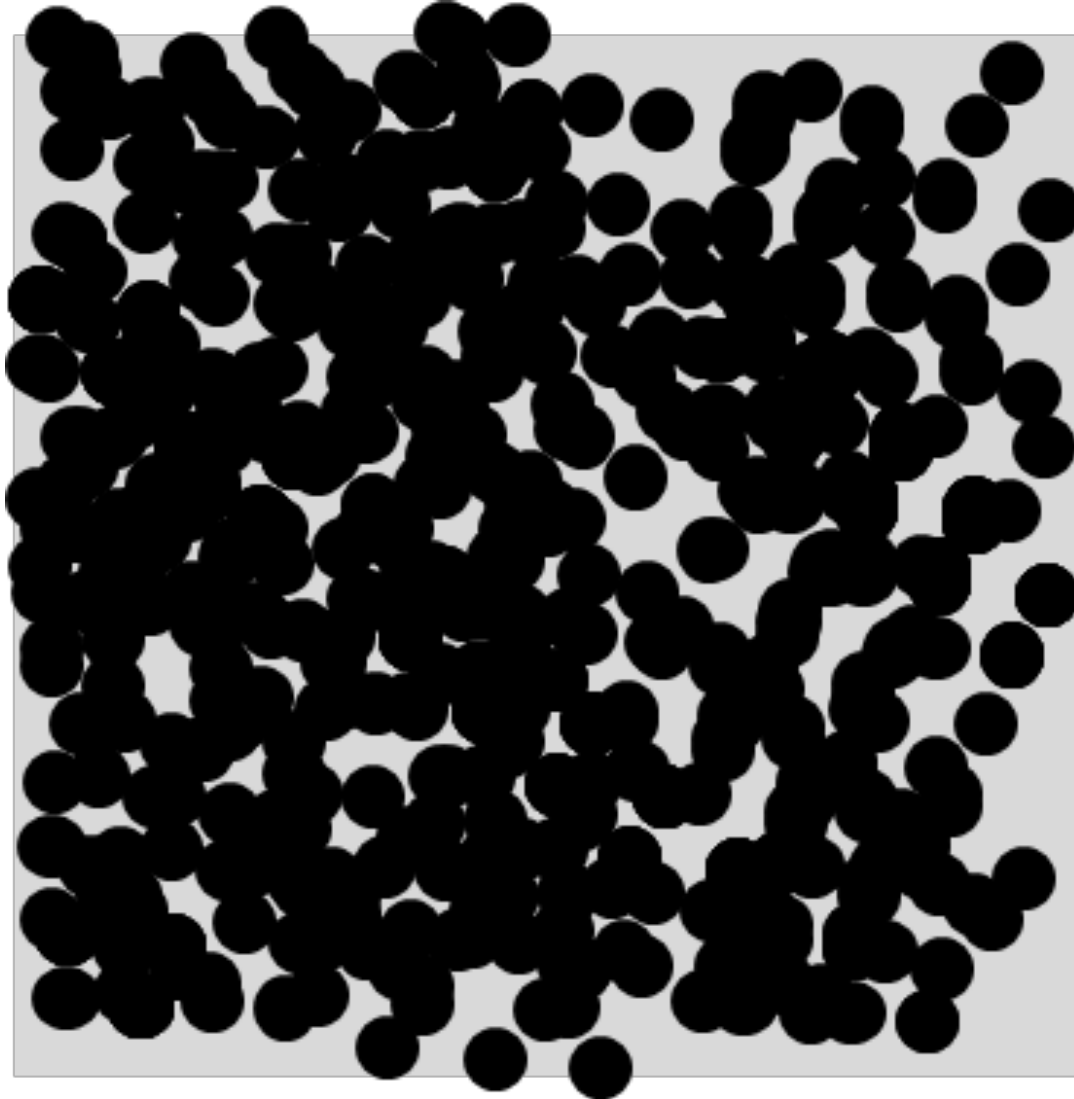


Identifying the position of the particle in different frames and obtaining the trajectory



Maximum displacement of the particle between frames

Identifying the position of the particle in different frames and obtaining the trajectory



GitHub Repository MOB.80

